

TSA Drone Challenge (UAV)

WTSA Spokane State Conference, 2026

Team ID: 21557-1

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Photo Log



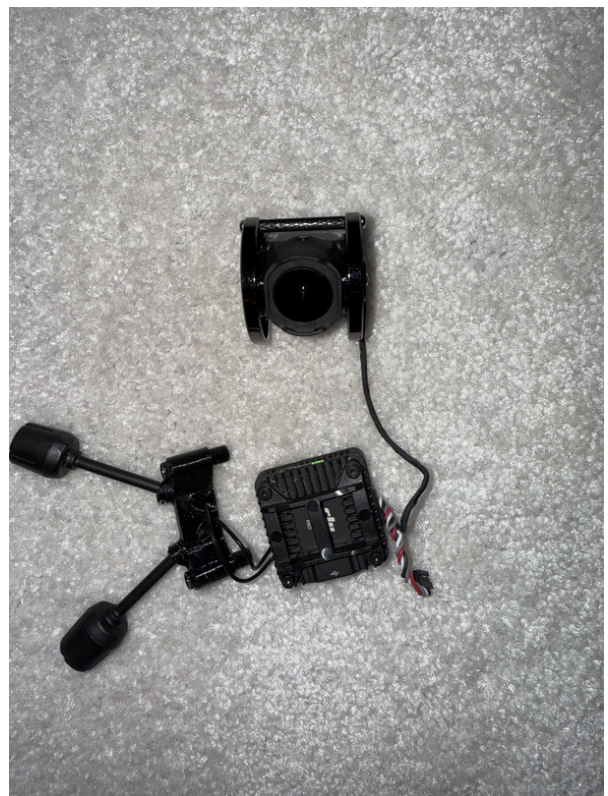
Gathering parts for the assembly of the frame



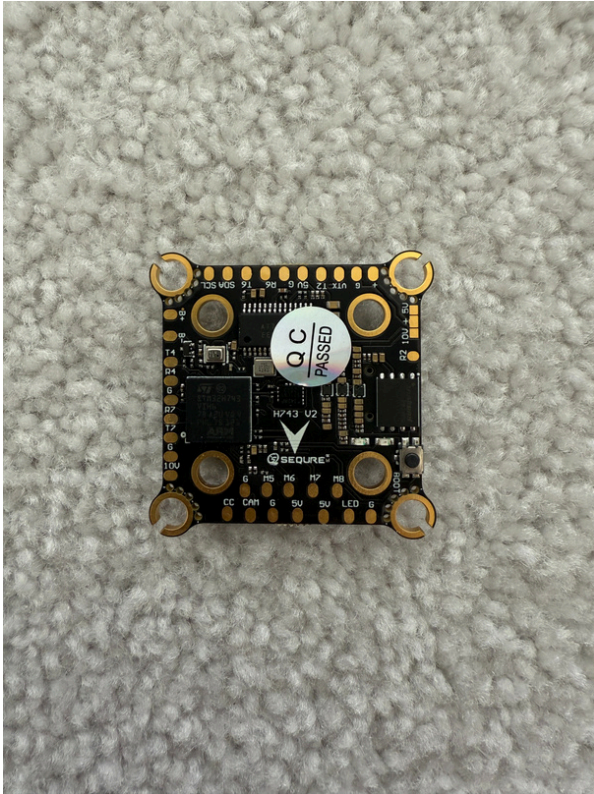
Assembled frame without top-plate



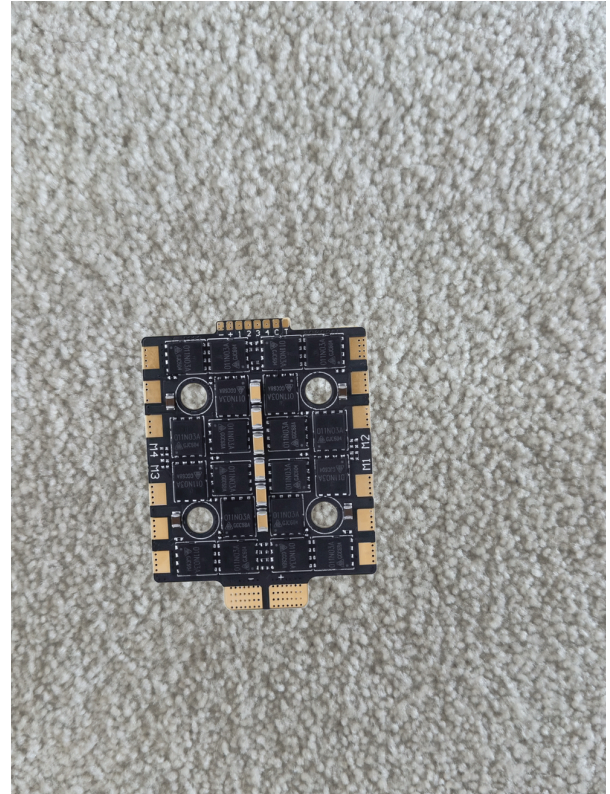
Dji O4 pro



Mounting of DJI O4 Pro in camera cage and antenna mounts



Flight Controller



Electronic Speed Controller (ESC)



4 Motors along with mounting screws



Propellers



DJI Goggles Integra, and Radio Master
Boxer



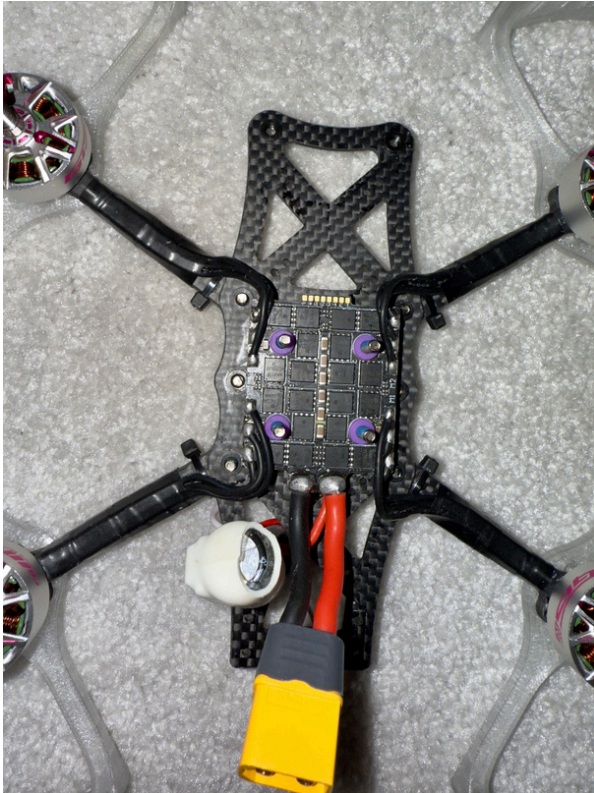
Claw assembly



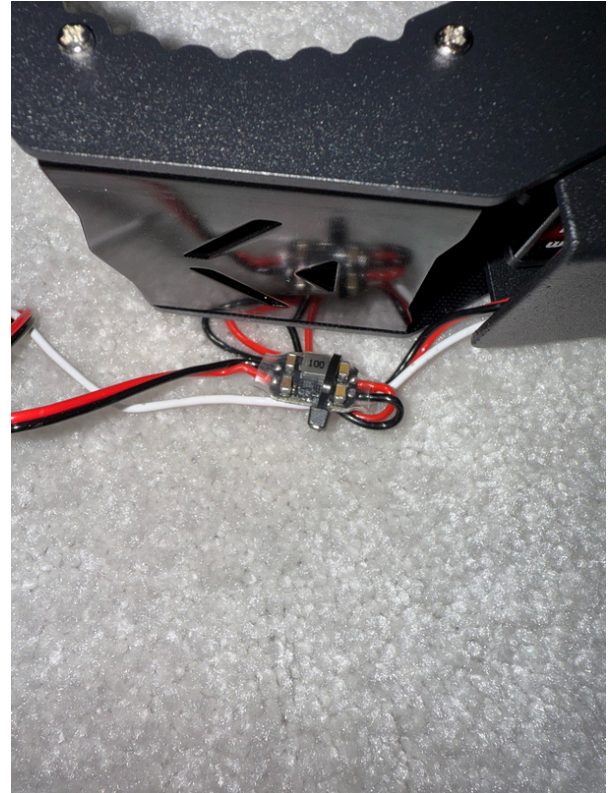
6S HV LIPO battery



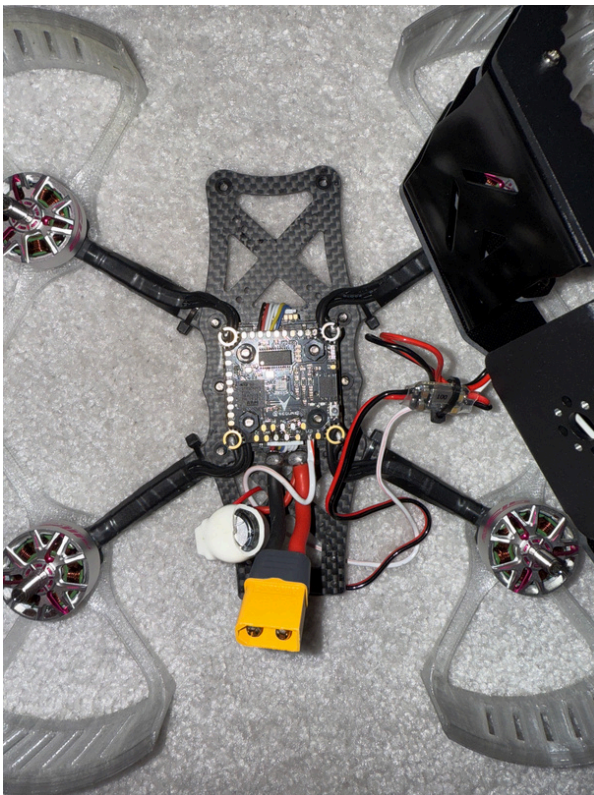
Happy Model EP1 and antenna mount



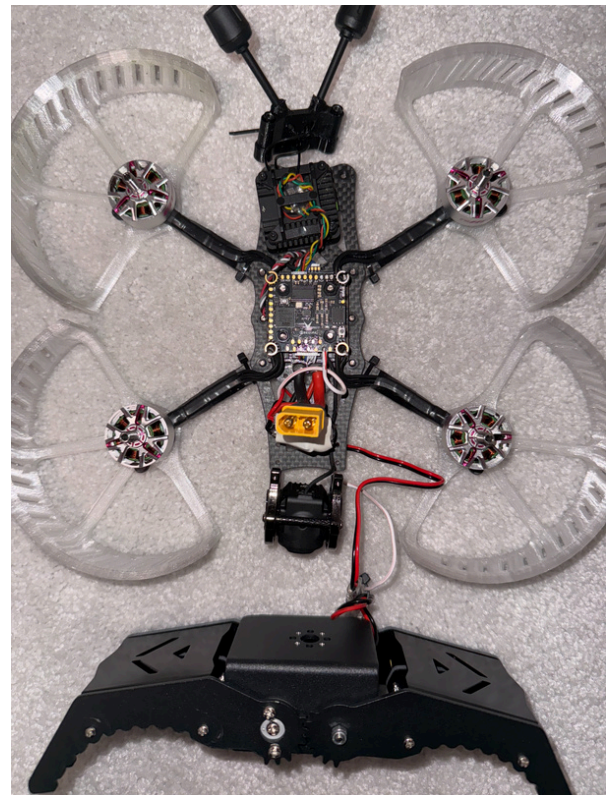
Soldering Battery Leads, Motor Wires, and Capacitors



Installation of Voltage regulator (BEC)



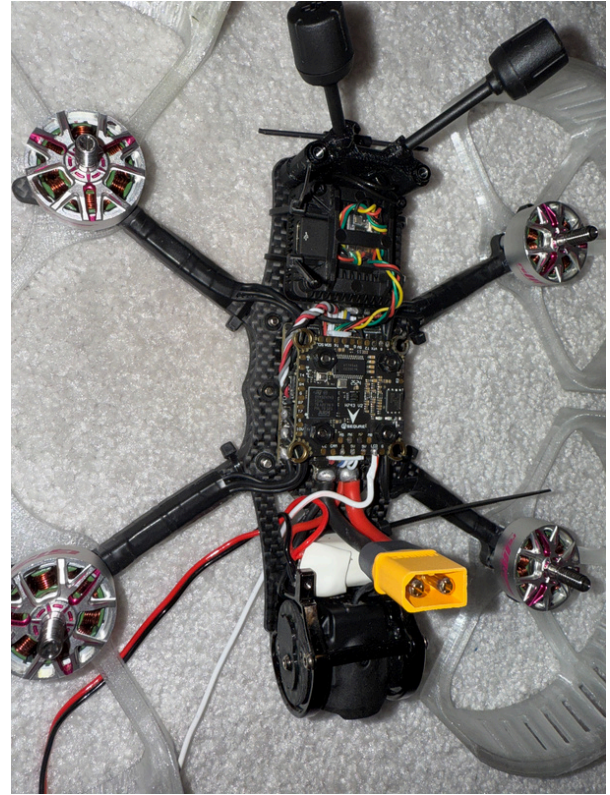
Soldering of control wire of servo to the flight controller + connecting ESC and FC



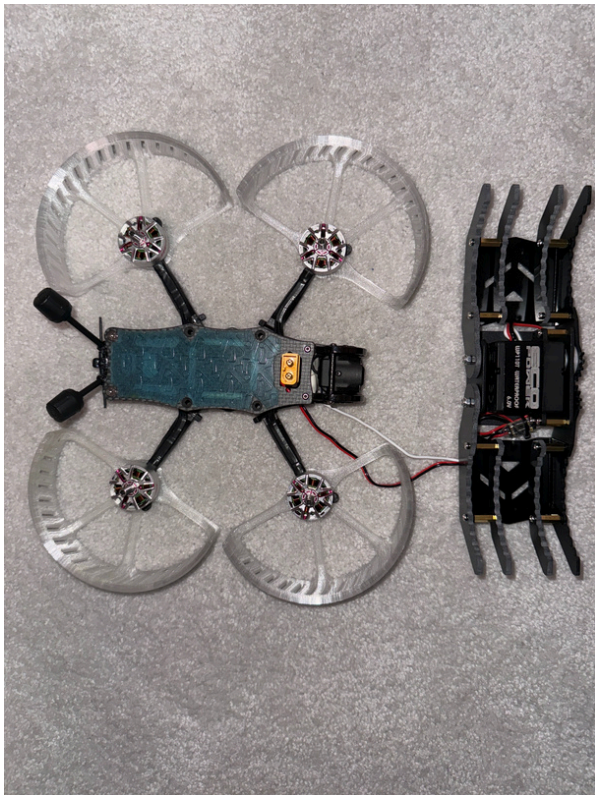
Plugging in DJI O4 Pro, and Happy Model EP1 receiver



Installation of camera cage to the frame



Mounting of DJI O4 Pro and Happy Model EP1



Securing Top-Plate to the drone frame



Positioning claw on frame and routing cables before installation



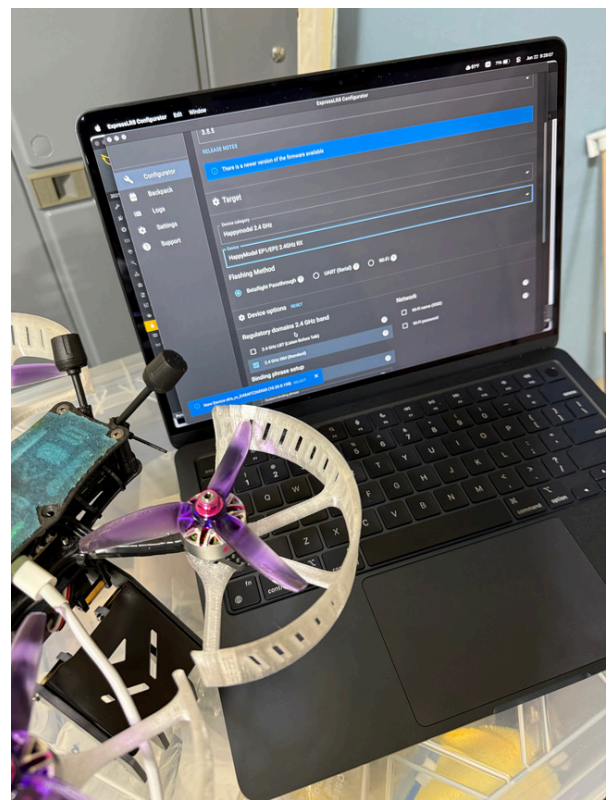
Securing Claw to frame with Zip Ties and managing servo cables



Installation of propellers



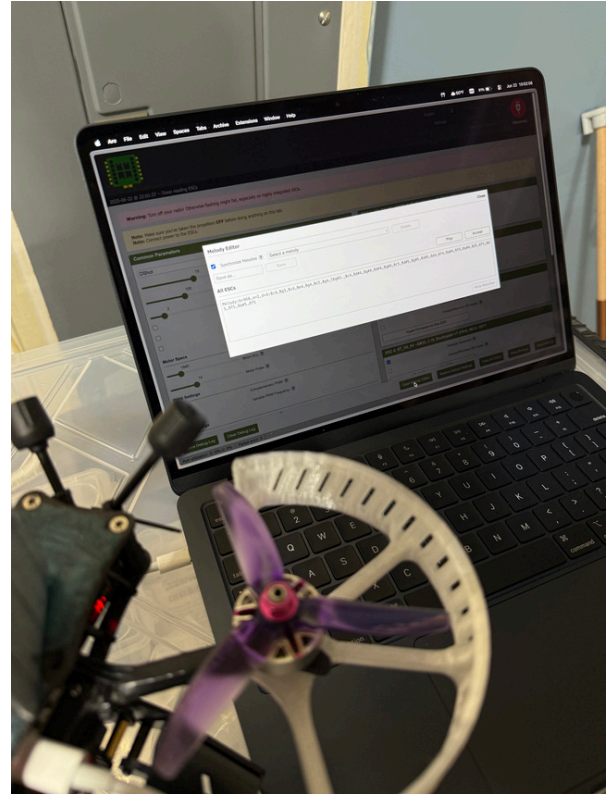
Adjusting and configuring drone with Betaflight (more details in software explanation)



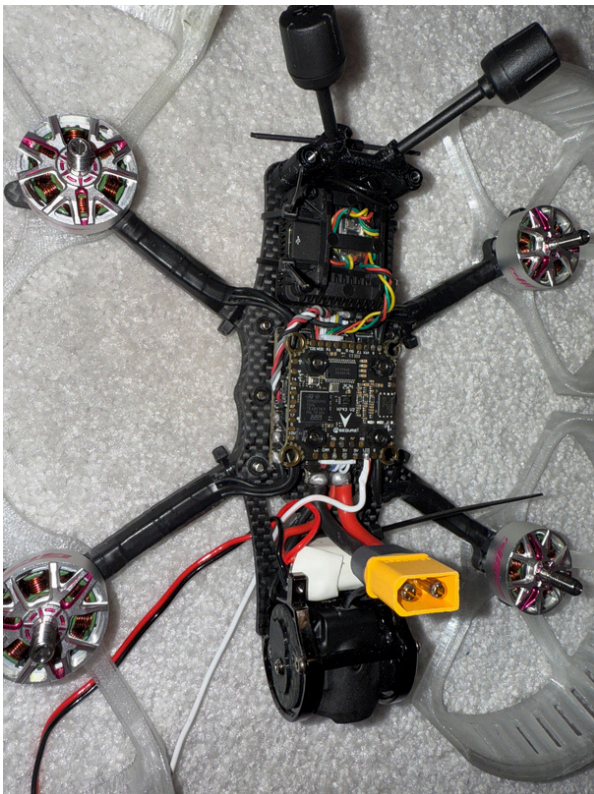
Flashing new firmware and binding code to the Happy model EP1 receiver using ExpressLRS Configurator.



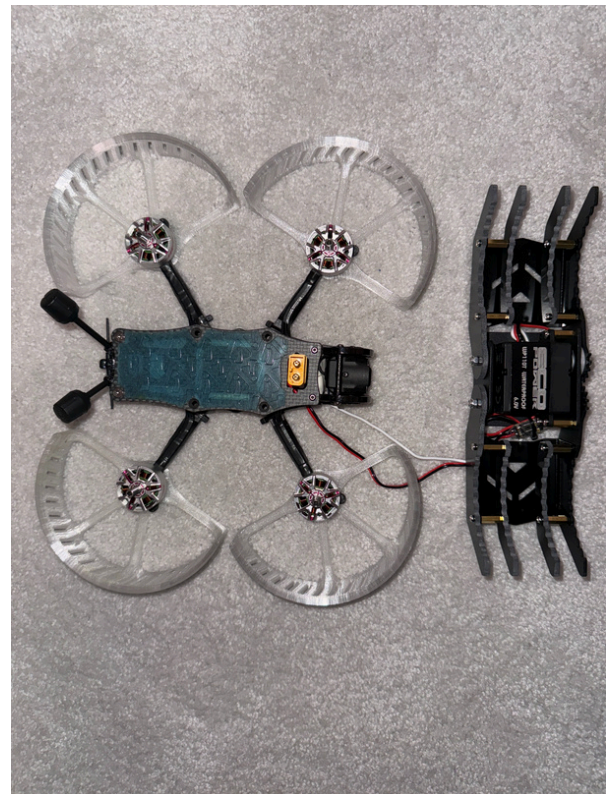
Updating ESC firmware to the latest version with AM32 Configurator



Changing ESC startup sound (for fun!)

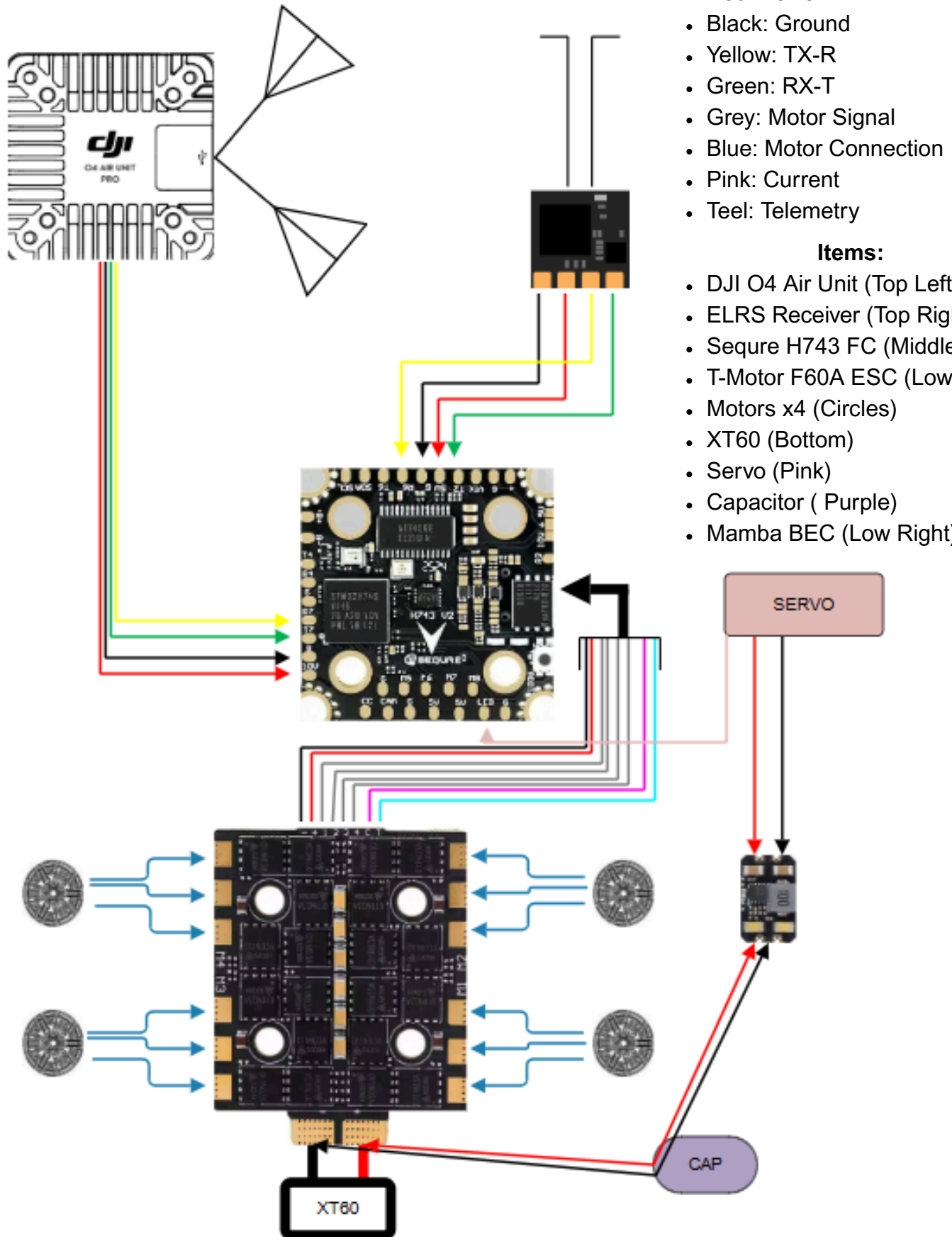


Body



Body

Wiring Diagram



Software Explanation

Section Overview:

We chose Betaflight as our flight controller software. Betaflight is a reliable open-source configurator with broad compatibility and is recommended by the manufacturer of our flight controller. Furthermore, we also used the AM32.ca website (AM32 configurator) to update the firmware and adjust the default configuration on our T-Motor ESC.

Along with this, we also used the ExpressLRS configurator which we used to update the firmware on our receiver and transmitter. Finally, we used DJI Assistant 2 to register and activate the DJI goggles and the DJI O4 module.

Betaflight:

- We used Betaflight for flight controller software, specifically to set up rates and to ensure that the precision of the drone is increased (e.g., reduced center sensitivity, increased expo for finer control when the drone is moving slowly, and we chose not to set a throttle limit since the drone is smaller compared to others, which means it has less lifting capacity).
- It was also used for remapping the LED pad on the flight controller to output servo commands to the servo so that it can be controlled from the controller.
- We used it for setting up the on-screen display (OSD), which provides the pilot with information about the battery voltage and an estimate of the flight time left before needing to land and switch out the battery.
- PID tuning was performed to allow the drone to fly without jittering. This was done by adjusting the gyro filter multiplier to reduce the amount of vibrations detected by the gyro to ensure smoother flight.
- Finally, we used it for reordering the motors to make sure that they spin the correct way using the built-in motor direction wizard in the Betaflight motor tab.

AM32 Configurator:

- We used the AM32.ca configurator as the ESC we selected to use on our drone utilized the AM32 firmware. Within the configurator, I set the motor KV setting to match the KV of the motor, as it reduces the amount of heat generated and improves efficiency.
- Along with this, I also used the AM32 configurator to set a motor tune so that when the ESC starts up, it plays a song of my choice.

ExpressLRS Configurator:

- Used to update the firmware on the Happy Model EP1 receiver on the drone, along with adding a binding phrase so that it cannot be connected to by unintended controllers.
- Also used to update the ExpressLRS module in my RadioMaster Boxer controller so that the receiver and drone communicate smoothly.

DJI Assistant 2:

- DJI Assistant 2 was used to set up and register both the DJI O4 PRO air unit and the DJI Goggles Integra.
- Also used to update firmware and ensure that the power settings and channel selections follow the rules and regulations for the local area.

PID Tuning:

PID tuning is a critical process in drone flight intended to ensure stability and responsiveness during operation. PID is an acronym for Proportional, Integral, and Derivative; it is a control loop feedback mechanism usually utilized by flight controllers. Each of these components plays a different role in how the drone reacts if it is off from some desired behavior:

1. Proportional (P): The part responsible for the existing error develops a corrective force proportional to the size of the error. For a drone, this means the flight controller adjusts the motor speeds depending on how much the position or angle of the drone is away from the setpoint. However, a gain that is too high will cause oscillations when the system overcompensates.
2. Integral (I): This part deals with the integral error over time. When a drone drifts to one side for reasons such as unevenly distributed weight or wind, the integral term tries to balance it out by progressively incrementing power from the other direction until it eliminates the drifting. Too much use of this variable leads to instability and slow reaction.
3. Derivative (D): This term is responsive to the rate of change of the error and thus dampens the system to prevent overshoot and oscillation. It smooths movements so that the drone does not oscillate or jerk unnecessarily during stabilization.

PID Tuning Applied to Our Drone:

PID tuning is performed to optimize our drone for stable, smooth flight and with reliable and agile control for complex maneuvers. By using Betaflight, we adjust the PID value for pitch, roll and yaw based upon the results of flight testing. This will allow us to ensure a reliable and agile drone for our complex maneuvers.

Tuning Process

1. Initial Testing: Begin with default PID settings, conducting hover and basic movement tests to detect instability or oscillations.
2. Proportional Gain Tuning: Gradually increase proportional gain for quick responses without oscillations. Reduce slightly if oscillations occur.
3. Integral Tuning: Adjusting the integral term to compensate for steady-state drift and provide a minor degree of lag but never ebbed instability.
4. Derivative Tuning: Change the derivative term to reduce oscillation, especially in situations with rapid changes in directions.

Why PID Tuning is Important:

Without proper tuning, a drone would never be able to remain stable in a hover. Instead, it would bob, drift, and likely crash if it were in flight. Additionally, tuning increases the drones responsiveness of the drone, making it easier for the pilot to fly precisely. Overall, changes of the PID values and testing their impact has been important, and successful.

Final PID Values:

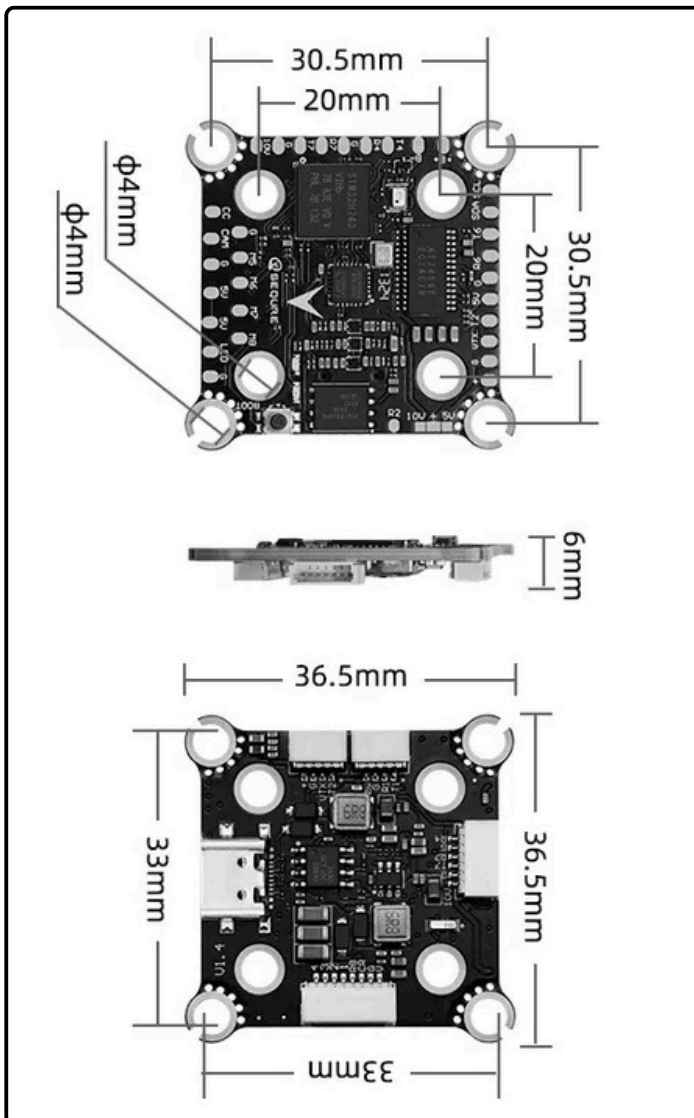
Axis	P	I	D	F
Roll	59	95	34	129
Pitch	49	79	30	108
Yaw	49	79	n/a	108

Other Settings:

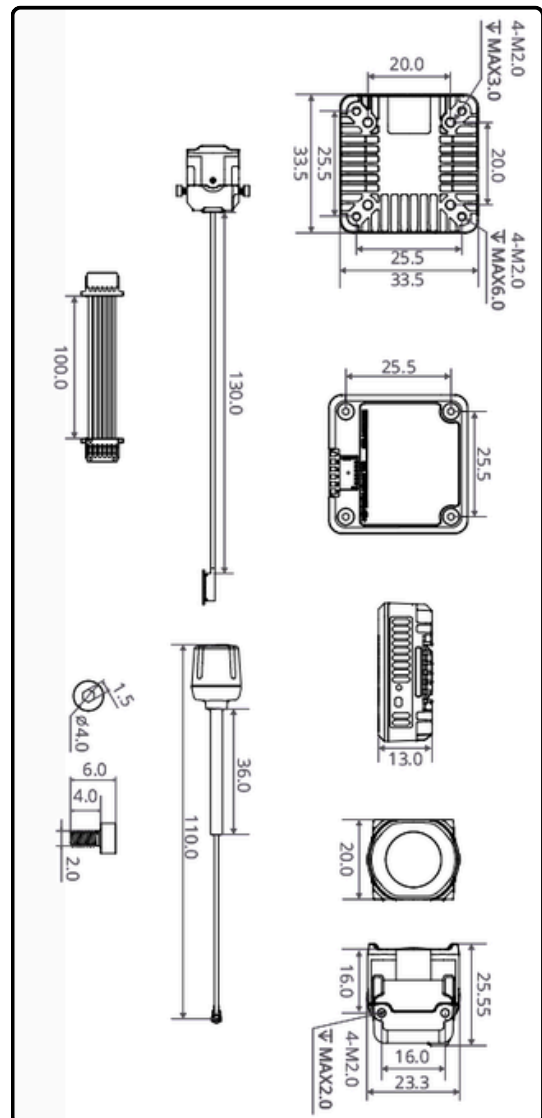
- Bidirectional DSHOT: With dshot_bidir ON, bidirectional DSHOT is enabled allowing the ESCs to send RPM telemetry back to the FC for RPM filtering.
- Angle Limit: angle_limit = 26 sets the maximum lean angle in Angle mode at 26 degrees to prevent excessive tilting.
- Feedforward Smoothing: Factor set to 45, and boost to 14, making response smooth and stick input more responsive.

Engineered Drawings

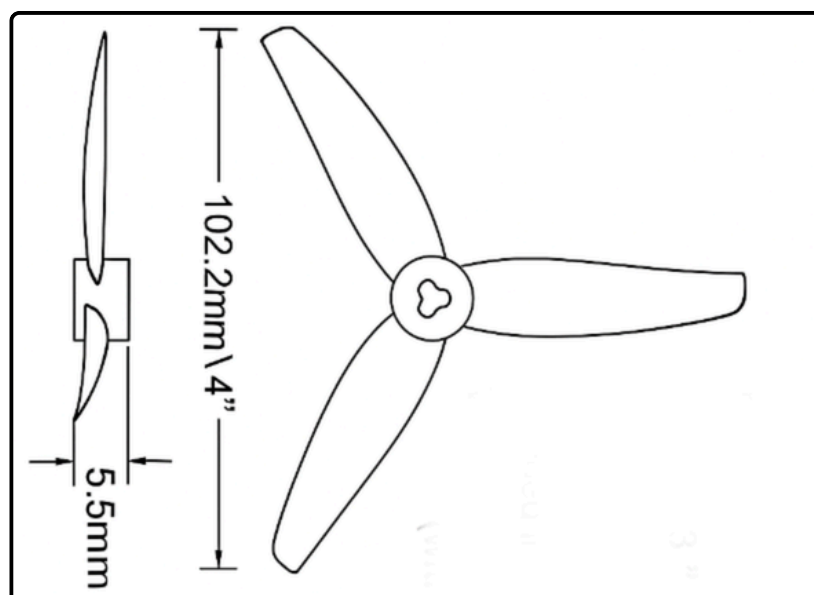
SEQURE H743



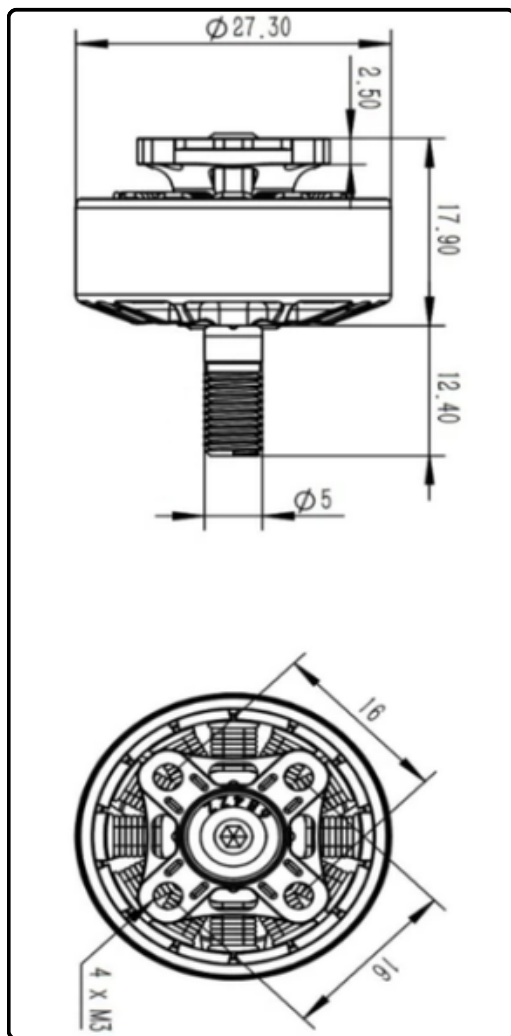
DJI O4 PRO



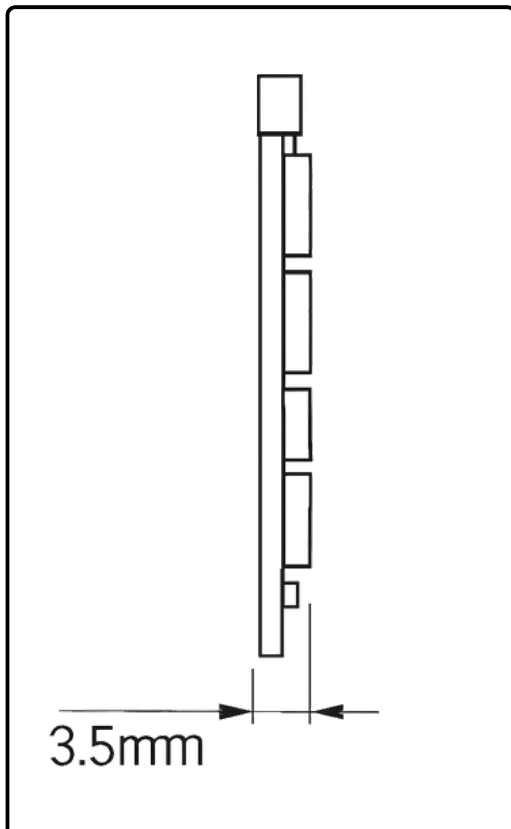
PROPELLERS



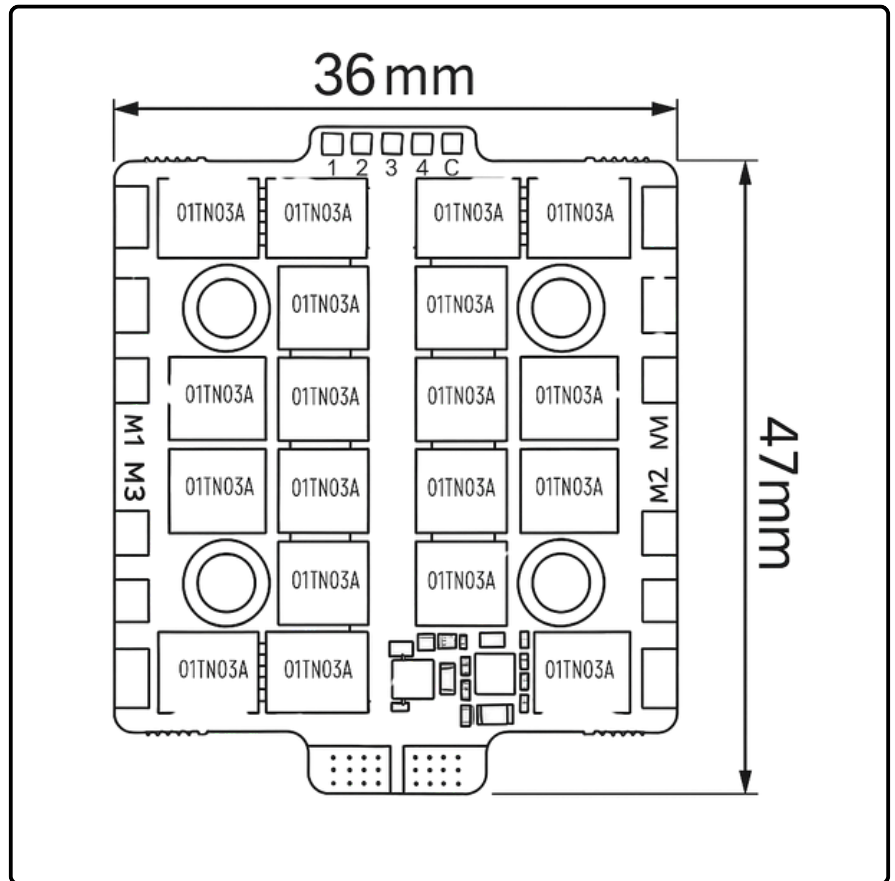
1960KV GTS V4 MOTORS



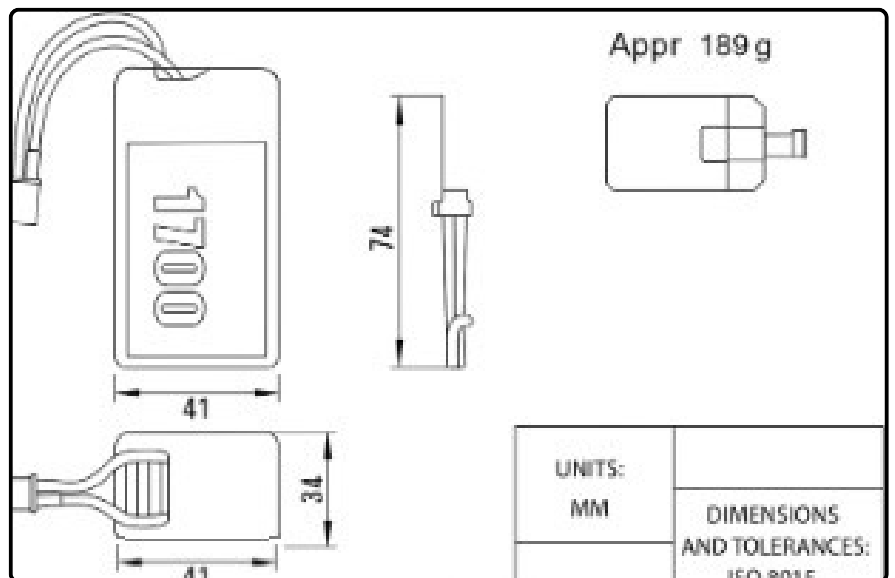
T-MOTOR F60A ESC WIDTH



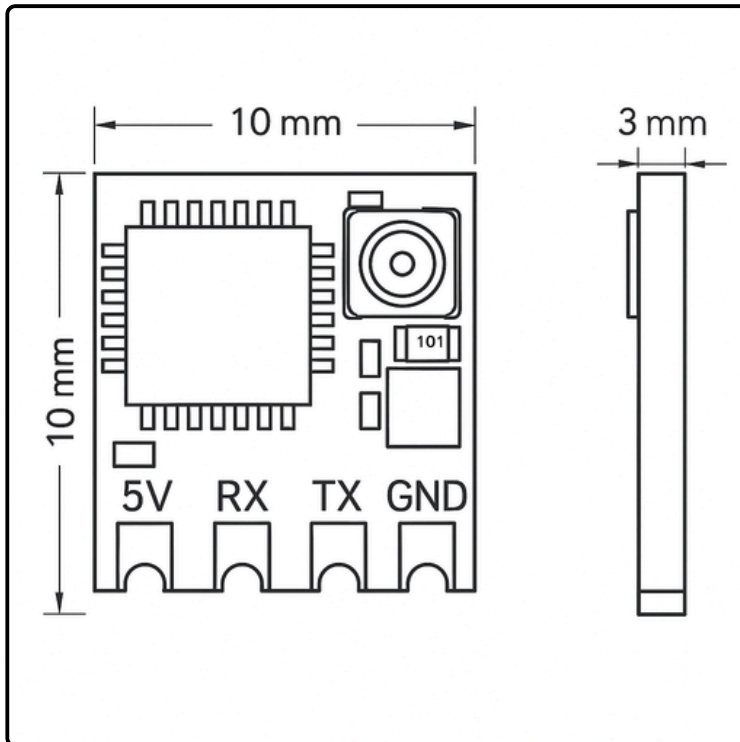
T-MOTOR F60A ESC



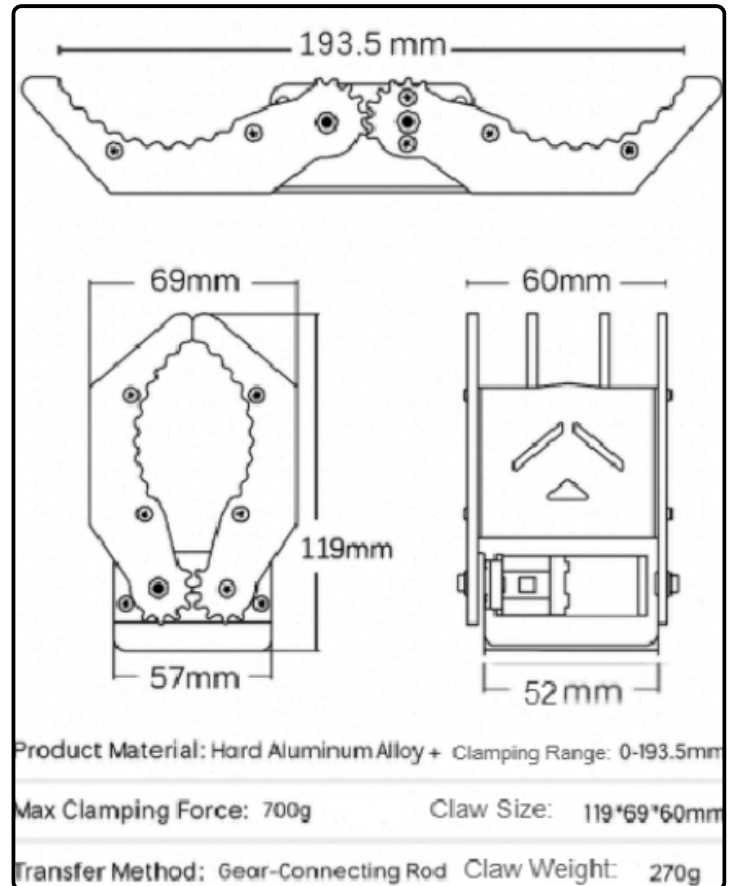
1700 MAH BATTERY



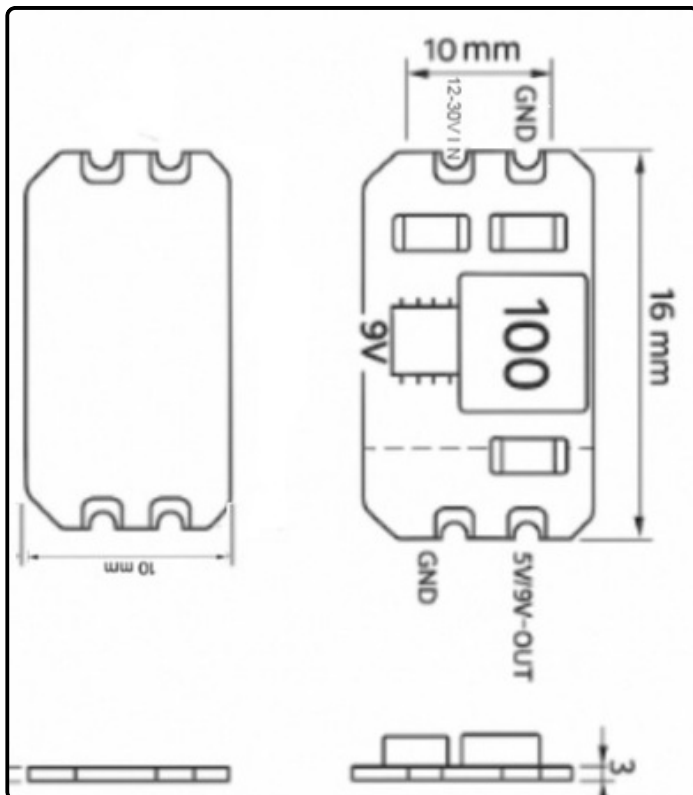
ELRS EP1 RX



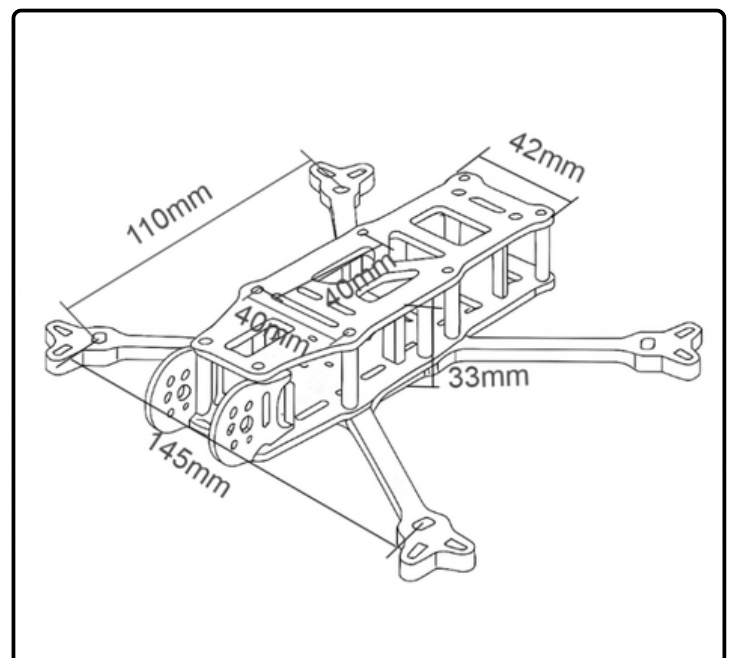
ALUMINUM SERVO CLAW



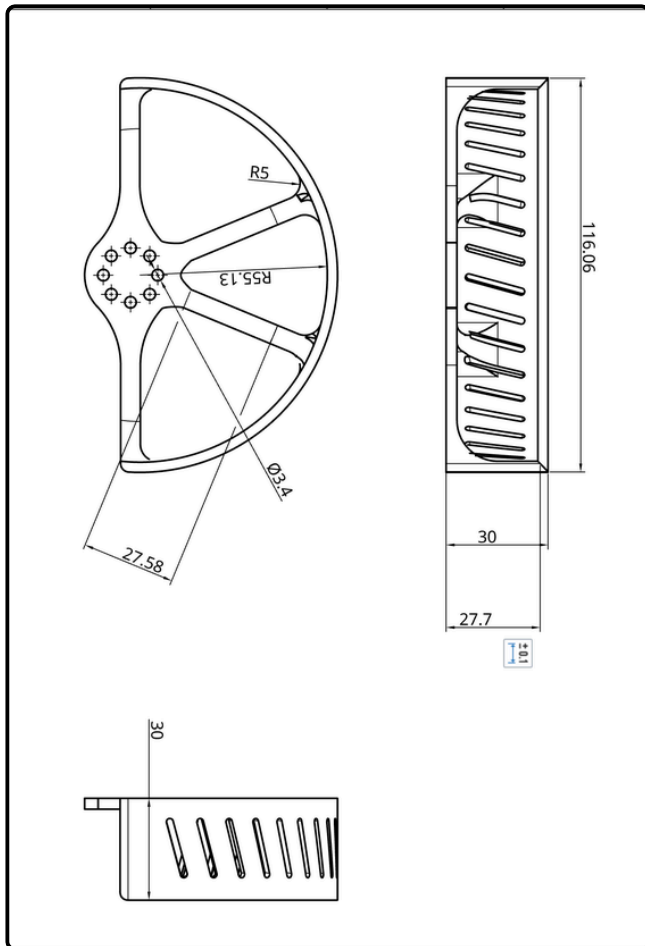
MAMBA BEC 5V/9V



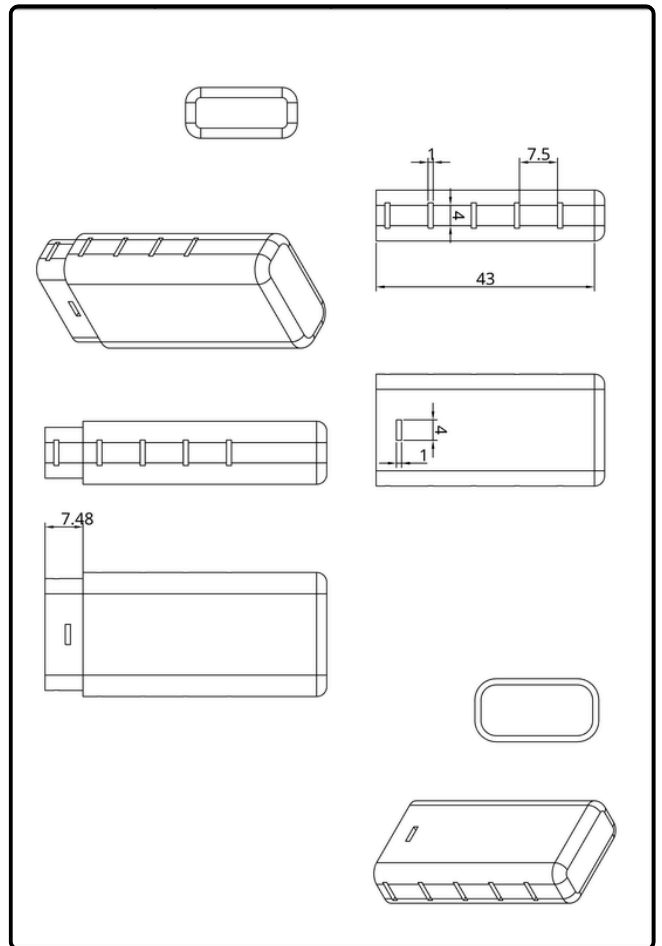
AOS 4 V5 FRAME



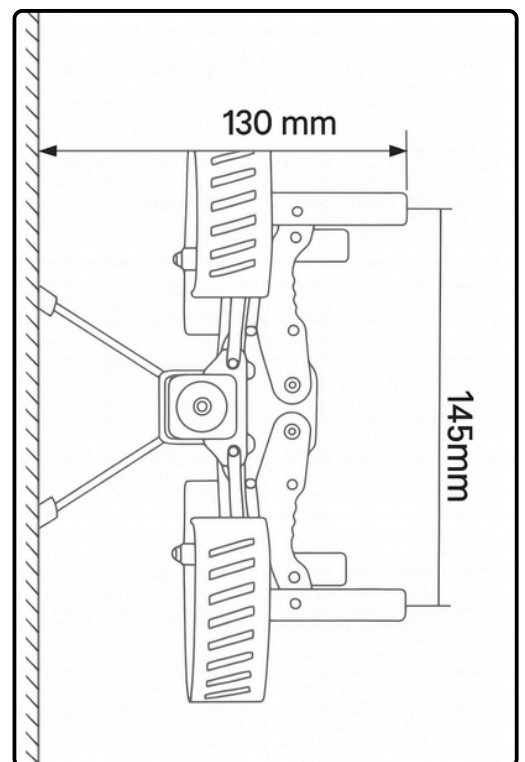
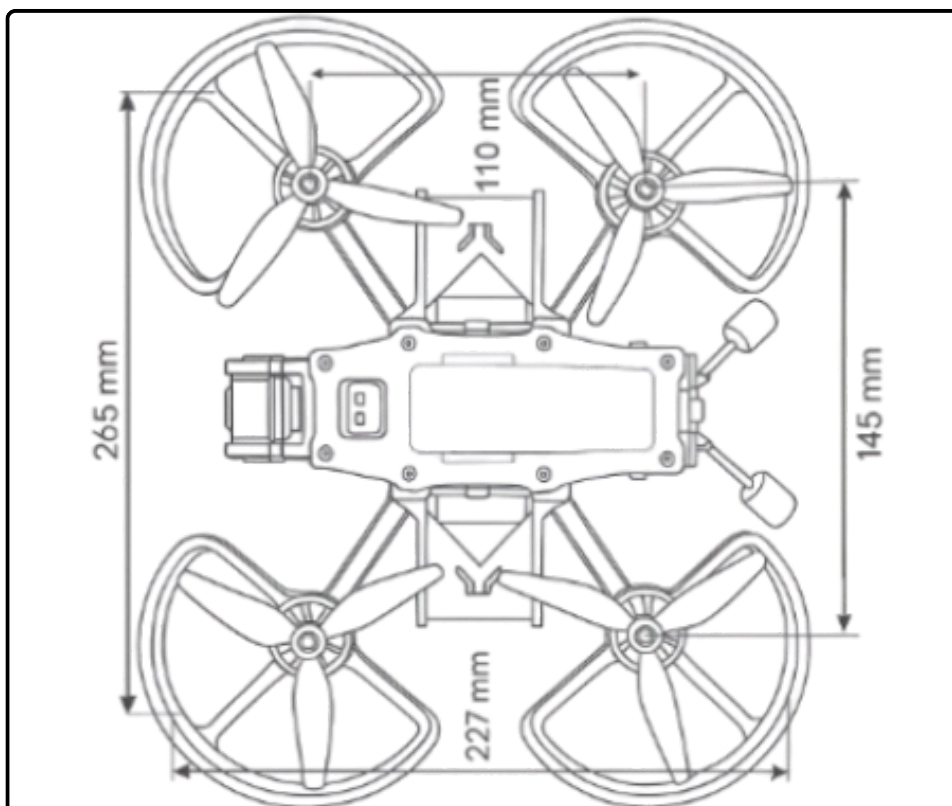
PROPELLER GUARDS (3D PRINTED)



ADJUSTABLE LANDING GEAR



FULL DRONE DRAWING TOP AND SIDE VIEW



Bill of Materials

Part Name	Link	Quantity	Cost (each)
iFlight AOS 4 V5 FPV Drone Frame Kit (Frame)	https://pyrodrone.com/products/iflight-aos-4-v5-fpv-drone-frame-kit	1	\$76.99
Sequare H743 V2 4-8S ICM42688P Dual BEC Flight Controller (Flight Controller)	https://pyrodrone.com/products/sequare-h743-v2-4-8s-icm42688p-dual-bec-flight-controller-20x20mm-30x30mm	1	\$54.99
Ummagrip Lite - Super Sticky Battery Pad (Battery Grip)	https://pyrodrone.com/products/copy-of-ummagrip-lite-super-sticky-battery-pad-choose-color?variant=40751361687595	1	\$4.59
HQ Prop 4x4.3x3 V1S (2CW+2CCW) PC (Propellers)	https://pyrodrone.com/products/hq-4x43x3-v1s-2cw-2ccw-pc-choose-color	8	\$3.99
DIATONE MAMBA MICRO 2A BEC (BEC voltage regulator)	https://www.aliexpress.us/item/3266806672319262.html?gatewayAdapt=glo2usa	1	\$3.87
RCinPOWER GTS V4 2207 1960Kv Motor (Motors)	https://www.racedayquads.com/products/rcinpower-gts-v4-2207-1960kv-motor-silver-pink	4	\$32.49
Tattu R-Line Version 5.0 1200mAh 22.2V 150C 6S1P Lipo Battery - XT60 (Batteries)	https://pyrodrone.com/products/tattu-r-line-version-5-0-1200mah-22-2v-150c-6s1p-lipo-battery-xt60?variant=40183673226269	2	\$32.99

Part Name	Link	Quantity	Cost (each)
GAONENG GNB LiPo Battery 6S 1550mAh 22.8V HV High Voltage 60C for 5 inch Long Range Drone FPV XT60 Connector (Batteries)	https://www.amazon.com/GAONENG-Battery-1700mAh-22-8V-Voltage/dp/B09MRF6LMY	2	\$42.99
DJI O4 Pro (Video System)	https://store.dji.com/product/dji-o4-air-unit-pro?vid=180741&from=pc-cart-title	1	\$229.00
Mechanical Claw Gripper Robotic Arm DIY (Claw)	https://www.amazon.com/Mechanical-Gripper-Building-Accessory-Replacement/dp/B0CMHSC9MY?th=1	2	\$38.99
T-MOTOR Hobby F60A mini 8S 4IN1 20X20 (ESC)	https://www.aliexpress.us/item/3266804682270526.html?gatewayAdapt=glo2usa	1	\$91.48
DJI Goggles Integra (FPV Goggles)	https://www.progressiverc.com/products/dji-goggles-integra	1	\$499.99
RadioMaster Boxer EdgeTX RC Transmitter (Controller)	https://www.racedayquads.com/products/pre-order-radiomaster-boxer-edgetx-rc-transmitter-choose-version?variant=40101052874865	1	\$167.99
Total	N/A	N/A	\$1,519.94

Drone Flight Regulations

Federal Aviation Administration (FAA) Regulations:

- **Part 107 (Small Unmanned Aircraft):** Operations that are not purely recreational require an FAA Remote Pilot Certificate. This may include drones flown in competition if it is non-recreational.
- **Recreational Flyers:** If the drone is only for recreation, pilots should do the following (mostly only if flying outdoors).
 - Pass the Recreational UAS Safety Test.
 - Register the drone with the FAA if it weighs 260g or more.
 - Fly as per safety guidelines.
 - Keep the drone within sight of the pilot and respect others' privacy.
 - Do not interfere with manned aircraft.
 - Do not fly near first responders.
 - Do not fly under the influence or be reckless/careless.
- **Airspace Restrictions:**
 - **Controlled Airspace:** Includes classes B, C, D, and E. Flights in space near airports like BNA need authorization through LAANC or FAA for Part 107 operations or with ATC for recreational use.
 - **Temporary Flight Restrictions:** Given for special events, presidential movement, or natural disasters. Drones are prohibited in these places.
 - **No-Fly Zones:** Permanent restrictions over areas like military bases, national parks, and infrastructure.
- **Altitude Limits:** Both recreational and Part 107 flights are limited to a maximum altitude of 400ft in uncontrolled airspace.
 - The FAA rules do not apply to non-commercial indoor use of drones
- **Washington State Regulations:**
 - WAC 172-110-070: Use of drones is prohibited in locations where there is a reasonable expectation of privacy, including restrooms, residential facilities, and dressing rooms.
 - RCW 9A.52 / RCW 9A.44: General state statutes regarding Trespassing and Voyeurism apply to drone operations; using a drone to record individuals in private settings without consent may lead to criminal charges.
 - WAC 352-32-130 (State Parks): Drones may only be flown in Washington State Parks with written permission (permit) from the director; operators must follow specific time and geographic restrictions.
 - WAC 220-413-070 (Wildlife): It is prohibited to use a drone to spot, locate, or report the location of wildlife for the purpose of hunting, or to harass any wild animal or bird.

Technology Student Association Regulations:

• Competition Regulations:

- Props off and battery disconnected when the UAV is out of the competition area.
- All drones in the pit area or outside the competition area must be powered off.
- All batteries and props must be inspected by the supervisor before flying.
- All drones must fly only in the competition tent, nowhere else.
- Prop guards are required.
- Any violation will result in disqualification.

• Drone Specifications:

- Must have 4 motors and 4 propellers, and propellers must be removable.
- All parts must be open-sourced, and batteries must be commercial Li-Po's.
- Propellers can range in size from 4 inches to 8 inches.
- The full drone must fit in an 18'x18' box.
- A camera and grippers are required to earn points in the competition.

• Safety:

- **Welding Blankets:** Teams are required to bring 2 welding blankets to cover the table and floor in their pit area.
- **Battery Charging:** Battery chargers and batteries, as they are being charged, must be placed on the welding blanket in the charging area.
- **Tools:** The use of tools with combustible fuel sources is prohibited.
- **Computers:** Computers must operate on battery power. Teams can bring a power bank.
- **Inspection:** All equipment, portfolio, tools, chargers, and computers must be arranged for inspection and safety check.

Resources

"A1." Bambu Lab Wiki, wiki.bambulab.com/en/a1. Accessed 26 October 2025.

Amazon.com: GAONENG GNB LiPo Battery 6S 1550mAh 22.8V HV High Voltage 60C for 5 Inch Long Range Drone FPV XT60 Connector: Toys and Games. www.amazon.com/GAONENG-Battery-1700mAh-22-8V-Voltage/dp/B09MRF6LMY. Accessed 2 Apr. 2026.

Amazon.com: Mechanical Claw Gripper Robotic Arm DIY Robot Building Accessory Replacement Parts (Black Mechanical Claw With Servo and Tester): Toys and Games. www.amazon.com/dp/B0CMHSC9MY?social_share=cm_sw_r_cso_sms_apin_dp_SKQWPQZA2XN4QAMJEG26. Accessed 26 October 2025.

Bardwell, Joshua. "FPV SHOPPING LIST | 4" Freestyle FPV | FPV-Know-All | Joshua Bardwell." FPV Know-It-All - Joshua Bardwell, 29 Mar. 2023, www.fpvknowitall.com/fpv-shopping-list-four-inch-freestyle. Accessed 26 January 2026.

"DIATONE MAMBA MICRO 2A BEC 5V/9V for DJI Air Unit - AliExpress 26." Aliexpress., www.aliexpress.us/item/3266806672319262.html?gatewayAdapt=glo2usa. Accessed 26 January 2026.

"HappyModel EP1 EP2 RX 2.4G ExpressLRS ELRS Nano Long Range Receiver for ES24TX Module RC Airplane FPV Long Range LR4 LR7 - AliExpress 26." Aliexpress., www.aliexpress.us/item/3266806674426813.html?gatewayAdapt=glo2usa. Accessed 26 January 2026.

Joshua Bardwell. "Betaflight 4.3 Servos Tab | COMPLETE WALKTHROUGH PART 14." YouTube, 26 June 2022, www.youtube.com/watch?v=L-6r2iX1p6s.
"---." YouTube, www.youtube.com/@JoshuaBardwell. Accessed 26 January 2026.

Oscar, and Oscar. "How to Setup DJI O4 Pro Air Unit and Best Settings for Camera/Transmission." Oscar Liang, 27 Apr. 2026, oscarliang.com/how-to-setup-dji-o4-pro. Accessed 26 January 2026.

Power and Battery Tab | Betaflight. betaflight.com/docs/wiki/configurator/power-tab. Accessed 26 January 2026.

"SEQUIRE H743 V2 ICM42688P FC 4-8S Dual BEC for X8 Racing Aerial Photography Freestyle." SEQUIRE, sequiremall.com/collections/fc-and-esc/products/sequire-h743-v2-fc-4-8s-dual-bec-x8-freestyle. Accessed 26 January 2026.

"T-motor F60a Mini 8s 4in1 20x20 Racing Esc for Fpv - Parts and Accs - AliExpress." Aliexpress., www.aliexpress.us/item/3266804682270526.html?gatewayAdapt=glo2usa. Accessed 26 June 2026.

RaceDayQuads. "Sequire H743 4-8S 20x20 / 30x30 Flight Controller - MPU6000." RaceDayQuads, n.d., <https://www.racedayquads.com/products/sequire-h743-4-8s-20x20-30x30-flight-controller-mpu6000>. Accessed 26 January 2026.

Work Log

Date	Task	Time Taken	Team member responsible (student initials)	Comments
10/26/25	Brainstorming part lists	3 hours	AB	We all spent some time discussing what parts we wanted to get and different options for each part we needed.
11/12/25	Compiling the final list.	1 hour	AB	We compiled the final parts list that we would get and made sure that we covered all of the parts that needed.
11/17/25	Ordering the parts.	0.5 hours	AB,DS	We purchased all of the parts.
12/3/25	Assembling the frame.	1.5 hours	AB	We got the frame and put up all the pegs and other parts necessary for future building on the frame.
12/16/25	Bolting the ESC on and soldering the motors.	3 hours	AB	The ESC and motors got delivered so we put that ESC onto the frame and soldered on the motors.
1/3/26	Attaching DJI Air Unit and RX onto the FC.	4 hours	AB	We attached the camera/air unit, and radio onto the flight controller so it could be viable for flying in the future and to work for competition.
1/12/26	Relocation of the capacitor.	3 hours	AB	We had to remove everything from the frame so we reroute wires to the front of the frame so we could place the capacitor there.

Date	Task	Time Taken	Team member responsible (student initials)	Comments
1/13/26	Attatching battery straps and organizing wires.	1 hour	AB,DS	We organized wires to reduce clutter and risk of motor colliding with wires.
1/15/26	Getting the camera to show up on external screen.	1.5 hours	AB	We had to download the VTX tables and get the VTX antenna at the right frequency with the camera.
3/5/26	Test Flight #1	0.5 hours	AB	We took the drone out to a nearRSy park and had a test flight to see stability.
3/5/26	PID Tuning #1	1.5 hours	AB	Learned how to PID tune and tuned to stabilize control and better flying. PID tuning is explained in more detail in previous section.
3/10/26	Test Flight #2	0.5 hours	AB,MB,DS	Had another test flight.
3/10/26	PID Tuning #2	0.5 hours	AB,MB	Tuned again to fix stability.
3/14/26	Test Flight #3	0.5 hours	AB,MB,DS	Had a final test flight.

Date	Task	Time Taken	Team member responsible (student initials)	Comments
3/14/26	PID Tuning #3	0.75 hours	AB,DS	Final PID tuning to fix all remaining stability issues.
3/20/26	Claw Attachment + First Pickup	3 hours	DS,MB	Modifying the claw design from last year on CAD
3/24/26	Testing and redoing the claw	4 hours	AB,MB,DS	Tested with the servo and redoing the claw since the servo only turned 90 degrees
4/1/26	Construct Practice Course	3 hours	AB,DS,MB	Built a replica of the course for Akash to practice flying.
4/2/26	Construct Backup Drone	4 hours	AB	Backup drone with the same parts in case something goes wrong.

Date	Task	Time Taken	Team member responsible (student initials)	Comments
3/14/26	PID Tuning #3	0.75 hours	AB,DS	Final PID tuning to fix all remaining stability issues.
3/20/26	Claw Attachment + First Pickup	3 hours	DS,MB	Modifying the claw design from last year on CAD
3/24/26	Testing and redoing the claw	4 hours	AB,MB,DS	Tested with the servo and redoing the claw since the servo only turned 90 degrees
4/1/26	Construct Practice Course	3 hours	AB,DS,MB	Built a replica of the course for Akash to practice flying.
4/2/26	Construct Backup Drone	4 hours	AB	Backup drone with the same parts in case something goes wrong.